

```
mysql> create database library_db;
Query OK, 1 row affected (0.01 sec)

mysql> use library_db;
Database changed
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn                | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565      | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084      | 1960           |
| 3       | 1984                 | 9780451524935      | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Members;SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name          | email                |
+-----+-----+-----+
| 101       | John Smith         | john.smith@email.com |
| 102       | Emma Wilson        | emma.wilson@email.com |
| 103       | Michael Brown      | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

+-----+-----+-----+
| member_id | full_name          | email                |
+-----+-----+-----+
| 101       | John Smith         | john.smith@email.com |
| 102       | Emma Wilson        | emma.wilson@email.com |
| 103       | Michael Brown      | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Loans (
  -> loan_id INT PRIMARY KEY,
  -> member_id INT,
  -> book_id INT,
  -> loan_date DATE,
  -> FOREIGN KEY (member_id) REFERENCES Members(member_id),
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

10 rows in set (0.00 sec)

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.01 sec)

```
mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> START TRANSACTION;
```

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
    -> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
    -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|      3 | 1984  | 9780451524935 |      1949      |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

```